

Python Quiz Game: Code Explanations and Learnings!

1. `print('...')` – This line of code displays all the text on the screen. Any text between the apostrophes or double apostrophes will be outputted as text when the user runs the programme.

- a. Example in my code:

```
print ('=====')  
print (' The Python Quiz')  
print ('=====')  
print ('Welcome!')
```

2. `Score = 0` – This line of code creates a variable which is a named box that stores a value. This value can later be changed through lines of code such as below.

- a. Example in my code:

```
score = 0  
score +=1
```

3. `input(), .strip(), .lower()` – This is an example of getting and cleaning a user input which I think is cool. Let's say the code is a question such as What is the capital of France and the answer is Paris, if the user were to input paris they would be wrong but if they used the code above AFTER the print function, it would accept it. In fact, it would accept any variation of the letters in Paris. For the code to work, the answer must not start with a capital letter as show below.

- a. Example in my code:

```
difficulty = input('What difficulty do you want to choose? Choose from  
Easy, Medium or Hard ').strip().lower()  
if answer == 'perfect':
```

4. `if, elif and else` – If checks a condition and if that condition is true the block of code runs. `==` is a comparison operator and checks if 2 values are equal. `elif`, which is short for else and if, is another condition checked only if the previous ones were **false**. `else` runs if neither if nor `elif` conditions are true.

- a. Example in my code:

```
if difficulty == 'easy':  
    ...  
elif difficulty == 'medium':  
    ...  
elif difficulty == 'hard':  
    ...  
else:  
    print('That is not a valid difficulty!')
```

5. `while true:` - This is a loop that keeps on going until the condition is true. The capital T true is a special value meaning always true, so while true keeps on going until you manually stop it. To break the loop, I utilize a break line of code at the end of each program that satisfies the true value.

- a. Example in my code:

```
while True:
    question = input('...').strip().lower()
    if question == 'bone marrow':
        ...
        break
    else:
        print('Its not that hard I swear, try again')
```

6. `Subject = input(...)` - Here I have switched topics by asking what subject they want next (code below) and then cleaning the input with the tools I talked about earlier `.strip().lower()` and then using `if subject == 'no': break` to exit the big loop and end the difficulty section.

- a. Example in my code:

```
subject = input('Would you like to do Chemistry or Maths next? Type  
Chemistry, Maths or No: ').strip().lower()
```

These are the main functions and key pieces of code that I have used in my python code!